MILITARY SPECIFICATION

ACTUATOR, PISTON, PA535
PARTS FOR
AND
LOADING, ASSEMBLING, AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication of parts, assembling, and packing for one type of piston actuator designated as Actuator, Piston, PA535.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-208 - Ink, Marking, Stencil, Opague (Porous and Non-Porous Surfaces)

MILITARY

MIL-P-116 - Preservation-Packing, Methods Of
MIL-B-117 - Bag, Sleeve and Tubing - Interior, Packing
MIL-P-22264 - Powder, Ignition, Gasless, A-1A

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1375
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1190 - Minimum Guidelines for Level C Preservation Packing and Marking

2.1.2 Other Government documents, drawings, and publication. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
12550906 - Actuator, Piston, PA535.

12550757 - Ballistic Environment Test Requirement of M762/M767 Components and Subassemblies.
12631059 - Actuator, Piston, PA535 Test Fixture Assembly.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-177 (latest revision)".)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (Dwg.) 12550906 and all associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Moisture content ignition powder. The moisture content at the time and place of loading the actuator shall not exceed 0.20 percent when tested as specified in 4.4.3.1.

3.4 Functioning tests - sensitivity and output.

3.4.1 No-fire test. The piston actuator, after being conditioned at $70^{\circ} \pm 10^{\circ}F$ for two hours minimum, (min.) shall not exhibit any signs of the output requirements of 3.4.2 after being exposed to the no-fire stimulus specified in 4.5.2.1.

3.4.2 All-fire output test. The piston actuator, after being conditioned at $70^{\circ} \pm 10^{\circ}F$ for two hours minimum, shall function on the attempt, shall shear a .050 inch diameter, aluminum pin, travel a minimum of .200 inch, and not shear the .080 inch diameter stop pin when tested as specified in 4.5.2.2 (See Dwg. 12631059)

3.4.3 No load output. The piston shall move out of the cup when fired in an uncontained mode. The header shall remain intact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

3.5 Environmental conditioning.

3.5.1 Resistance between electrode and cup, thermal shock, vibration and function. The piston actuator shall comply with the resistance requirement of Dwg. 12550906, with the output requirements of 3.4.2 and shall function on the first attempt after being exposed to thermal shock and vibration when tested as specified in 4.5.3.1.

3.5.2 Waterproofness. The actuator shall comply with the output requirements of 3.4.2 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70° ± 10°F when tested as specified in 4.5.3.2.

3.5.3 Cold temperature. The piston actuator assembly shall comply with the output requirements of 3.4.2 after being conditioned at minus 65° ± 5°F for two hours minimum and tested as specified in 4.5.3.3.

3.5.4 Hot temperature. The piston actuator assembly shall comply with the output requirements of 3.4.2 after being conditioned at 160° ± 5°F for two hours minimum and tested as specified in 4.5.3.4.

3.5.5 Shock G. The piston actuator assembly shall comply with the output requirements of 3.4.2 after being subjected to a 30000g setback force as specified in 4.5.3.5. The piston actuator assembly shall be conditioned at 70° ± 10°F for 2 hours minimum prior to testing.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples shall be as specified in the contract.

3.7 Change of qualified product. After first article, the manufacturer shall notify the purchaser prior to the implementation of any change which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirement. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the purchaser for review. Based upon the review the purchaser will approve or disapprove the change and notify the manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser. Changes representative of those which are subject to this requirement are any design changes in the manufacturing processing such as:

a. Package structure and material changes or changes to the geometry.

b. Changes to process flow charts, assembly flow chart and testing flow charts.

c. Conductor materials.

d. Assembly operations move from one line or building to another.

3.8 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner, and all manufacturing, processing, and assembling operations shall be correctly performed. All parts and assemblies shall be clean, free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All required markings and stampings shall be neat and sharply defined. Required packing shall be dry and shall pass qualification test as specified in 4.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in the quantities as indicated:

PART DESCRIPTION	DRAWINGS	QUANTITY
Cup	12631048	25
Bridge Plug and Ignition Assy	12550782	25
Plug Assembly	12550784	25
Header	12631057	25
Piston	12631050	25
Actuator, Piston, PA535	12550906	250

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I as specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.3.4 Component qualification. Qualification of the component supplied to this specification is not complete until it has been submitted to the passed additional tests performed by the purchaser. These tests shall include 30,000 G acceleration tests, 30,000 RPM spin tests and aging tests to evaluate long term storage characteristics. Coordination of testing and test reporting shall be the responsibility of the purchaser.

4.3.4.1 Re-qualification. Re-qualification of changed component(s) may depend on the results of circuit and application analysis conducted as a result of change(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER See Below
	PA535 Piston Actuator and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Actuator, Piston, PA535 (Dwg. 12550906)	250			
	Examination for defects	1/		3.2	4.4.2.7
	Functioning tests-sensitivity and output				
	a) No-fire test	1/	0-1	3.4.1	4.5.2.1
	b) All-fire output test	75, 2/	0-1	3.4.2	4.5.2.2
	Environmental tests				
	a) Resistance between electrode & cup, thermal shock, vibration, & function	50, 2/	0-1	3.5.1	4.5.3.1
	b) Waterproofness test	25, 2/	0-1	3.5.2	4.5.2.2
	c) Cold test	25, 2/	0-1	3.5.3	4.5.3.3
	d) Hot test	25, 2/	0-1	3.5.4	4.5.3.4
	e) Shock, G Test	25, 2/	0-1	3.5.5	4.5.3.5
	No Load Test	25	0-1	3.4.3	4.5.2.3

NOTES: 1/ The entire quantity of piston actuators shall be subjected to the specific examination/test prior to testing.

2/ Samples for this test shall be taken from no-fire test samples.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
	PA535 Piston Actuator and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg. 12631048)				
	Examination for defects	25	0-1	3.2	4.4.2.1
	Bridge Plug and Ignition Assembly (Dwg. 12550782)				
	Examination for defects	25	0-1	3.2	4.4.2.2
	Plug Assembly (Dwg. 12550784)				
	Examination for defects	25	0-1	3.2	4.4.2.3
	Insulation resistance between electrode and header of plug assembly	25	0-1	3.2	4.5.5
	Pull test of electrode	25	0-1	3.2	4.5.8
	Header (Dwg. 12631057)				
	Examination for defects	25	0-1	3.2	4.4.2.4
	Piston (Dwg. 12631050)				
	Examination for defects	25	0-1	3.2	4.4.2.5

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12631048
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY 12550906
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Insider diameter, min.		0.40%	3.2	Gage
102	True position of electrode hole incorrect		0.40%	3.2	Gage
103	Overall length		0.40%	3.2	Gage
104	Cup dented, distorted, or otherwise damaged		0.40%	3.2	Visual
Minor					
201	Outside diameter (O.D.)		0.65%	3.2	Gage
202	Evidence of Poor Workmanship		0.65%	3.8	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.2	Bridged Plug and Ignition Assembly		1 1	12550782
				NEXT HIGHER ASSEMBLY 12550906

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spot charge missing		0.40%	3.2	Visual
102	Cracks or voids in spot charge		0.40%	3.2	Visual
103	LMNR composition missing or does not cover spot charge completely		0.40%	3.2	Visual
104	Ignition material on outside of plug		100%	3.2	Visual
Minor					
201	Evidence of Poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550784
4.4.2.3	Plug Assembly				NEXT HIGHER ASSEMBLY 12550783
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull test of electrode	50	0-1	3.2	4.5.8
102	Insulation resistance test		100%	3.2	4.5.8
103	Outside diameter		0.40%	3.2	Gage
104	Plating improper	1/	1/	3.2	4.5.7
105	Glass pellet cracked		0.40%	3.2	Visual
106	Length of exposed pellet, max		0.40%	3.2	Gage
Minor					
201	Evidence of Poor workmanship		0.65%	3.8	Visual

NOTES:

1/ Sampling and rejection shall be in accordance with MIL-G-45204.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12631057
4.4.2.4	Header				NEXT HIGHER ASSEMBLY 12550784
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Small inside diameter		0.40%	3.2	Gage
102	True position of large I.D.		0.40%	3.2	Gage
103	True position of small outside diameter		0.40%	3.2	Gage
104	Total length		0.40%	3.2	Gage
105	Length of small O.D.		0.40%	3.2	Gage
Minor					
201	Large O.D.		0.65%	3.2	Gage
202	Small O.D.		0.65%	3.2	Gage
203	Evidence of Poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12631050
4.4.2.5	Piston				NEXT HIGHER ASSEMBLY 12631049
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length		0.40%	3.2	Gage
102	Length of cavity		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Inside diameter		0.40%	3.2	Gage
105	Surface finish of outside diameter improper		0.40%	3.2	Gage
Minor					
201	True position of inside diameter		0.65%	3.2	Gage
202	Evidence of Poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70827 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12550906
4.4.2.6	Actuator, Piston, PA535				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup cracked or split		100%	3.2	Visual
2	Explosive on exterior of cup		100%	3.2	Visual
Major					
101	Length of electrode		0.40%	3.2	Gage
102	Length, sealed end to base		0.40%	3.2	Gage
103	Diameter of cup		0.40%	3.2	Gage
104	Resistance between electrode and cup		0.40%	3.2	Gage, 4.5.6
105	Sealant not present 360°		100%	3.2	Visual
106	Presence of sealant on electrode beyond sealed area		0.40%	3.2	Visual
107	Sensitivity - no-fire	200	0-1	3.4.1	4.5.2.1
108	Functioning - all-fire	2/	0-1 1/	3.4.2	4.5.2.2
109	Environmental test				
	a) Resistance between electrode and cup, thermal shock, vibration and function	50	0-1	3.5.1	4.5.3.1
	b) Waterproofness	25	0-1	3.5.2	4.5.3.2
	c) Cold temperature conditioning	25	0-1	3.5.3	4.5.3.3
	d) Hot temperature conditioning	25	0-1	3.5.4	4.5.3.4
110	No load test	25	0-1	3.4.3	4.5.2.3
	Shock, G test	25	0-1	3.5.5	4.5.3.5
Minor					
201	Evidence of Poor workmanship		0.65%	3.8	Visual

NOTES:

1/ Any actuator that fails to function on first attempt or fails to produce the required output shall be classed defective.

2/ Samples from this test shall be taken from no-fire samples.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the Qualification Inspection and Quality Conformance Inspection Tables and as specified in 4.4.3.1.

4.4.3.1 Moisture content of ignition powder. The contractor shall provide adequate controls to ensure compliance with the requirement of 3.3 and shall test for verification one (1) sample of material from each eight (8) hours of production in accordance with 4.5.1. Composite samples shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of material represented by the sample shall be rejected. If the piston actuators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all piston actuators loaded with the non-conforming material, shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Moisture content ignition powder. The moisture content shall be determined in accordance with MIL-P-22264. Observation shall be made for compliance with the applicable requirement.

4.5.2 Functioning tests - sensitivity and output.

4.5.2.1 No-fire test. The piston actuator, maintained at the conditioning temperature, shall be subjected to the specified no-fire stimulus in accordance with the requirements of Dwg. 12550906. Any sample which fails to comply with the requirement of 3.4.1 shall be classed defective and shall cause the lot to be rejected. (Non-destructive test).

4.5.2.2 All-fire output. The piston actuator, maintained at the conditioning temperature, shall be assembled in the specified test fixture, also maintained at the conditioning temperature, (Dwg. 12631059) and subjected to the specified all-fire stimulus in accordance with the requirements of Dwg. 12550906. Observation shall be made for compliance with the requirements of 3.4.2. Any sample which fails to fire on the first attempt shall be classed defective. (Destructive test).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

4.5.2.3 No load output. The actuator shall be positioned unconstrained in a test fixture and discharged. The piston shall leave the cup and the header remain in the cup.

4.5.3 Environmental tests.

4.5.3.1 Resistance between electrode and cup, thermal shock, vibration, and function. The piston actuators shall be tested as follows, in the sequence specified, using approved equipment:

a. Measure resistance between the electrode and cup in accordance with 4.5.6.

b. Condition for two (2) hours minimum at minus 65° ± 5°F.

c. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 160° ± 5°F.

d. Condition for two (2) hours minimum at 160° ± 5°F and 20% maximum relative humidity.

e. Remove from hot chamber and immediately transfer to the vibrator.

f. Vibrate for 15 minutes, minimum, in a sweep covering a frequency range of 25-500-25 cycles per second, parallel to the transverse orthogonal axis, with output end down and a 2 ± 0.2 G's peak amplitude.

g. Measure resistance between electrode and cup in accordance with 4.5.6.

h. The piston actuators shall be tested as specified 4.5.2.2. Any piston actuator which fails to comply with the applicable requirement shall be classed defective. (Destructive test).

4.5.3.2 Waterproofness. The piston actuators shall be immersed in water at a two (2) to three (3) inch depth for 48 hours, minimum at the conditioning temperature. At the end of the immersion time, the piston actuators shall be removed from the water, tested according to 4.5.2.2, and observed for compliance with the requirements of 3.4.2. Any sample which fails to comply with the applicable requirements shall be classed defective. (Destructive test).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

4.5.3.3 Cold temperature. The piston actuator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.2.2 (Destructive test).

4.5.3.4 Hot temperature. The piston actuator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.2.2 (Destructive test).

4.5.3.5 Shock, G. The piston actuator assemblies, maintained at the conditioning temperature, shall be subjected to a 30,000g setback force in accordance with the ballistic environment test requirements of M762/M767 components and subassemblies, 12550757. After completion of the test, the piston actuator assemblies shall be tested as specified in 4.5.6 (destructive test).

4.5.4 Resistance between electrode and header. The bridge assembly shall be connected to the equipment and the bridge resistance shall be measured when a current limited to one (1) milliampere is applied. Any bridge assembly that fails to comply with the drawing requirement shall be classed defective and removed from the lot. (Non-destructive test).

4.5.5 Insulation resistance between electrode and header of plug assembly. The required D.C voltage shall be applied to the plug assembly between the electrode and the header for not less than one or more than two seconds. If the resistance measures less than the specified resistance, the plug assembly shall be classed defective and removed from the lot. (Non-destructive test).

4.5.6 Resistance between electrode and cup. The piston actuator assembly shall be connected to the equipment and the resistance shall be measured between the electrode and the cup when a current limited to ten (10) milliampere is applied. Any actuator assembly that fails to comply with the drawing requirement shall be classed defective and removed from the lot. (Non-destructive test).

4.5.7 Plating of electrodes. The plating shall be verified with the applicable drawing requirement. (Test samples shall not be returned to the lot).

4.5.8 Pull test of electrode of plug assembly. The plug assembly shall be tested in accordance with the applicable drawing requirements. (Destructive test).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

5 PACKAGING

5.1 Unit Packaging

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Actuators, Piston shall be packaged in accordance with MIL-STD-1190 and the applicable section (s) of Code of Federal Regulations, Title 49. In addition, each piston actuator shall be electrically shorted during shipment and storage by means of a metallic connection between electrode and cup.

Unless other packaging is used to provide barrier protection, the following instructions apply:

Each intermediate fiberboard box, set-up box or comparable container, after blunting corners, shall be placed in a water and vapore proof bag manufactured in accordance with Type I, Class E of MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the Manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C. A suitable number of Actuators Piston packaged as specified in 5.1.3, shall be packed in an outer box in accordance with MIL-STD-1190 and the applicable section(s) of Code of Federal Regulations, Title 49.

5.3 Marking.

5.3.1 Level A. Not applicable.

5.3.2 Level B. Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

5.3.3 Level C. Each inner container and intermediate fiberboard box, set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped as required by the applicable section(s) of Code of Federal Regulations, Title 49, and in accordance with MIL-STD-129, and the following information:

a. (Number) Actuator, Piston, PA535: Lot Number.

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4" high applied with black waterproof ink. Stencil ink shall conform to A-A-208.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Dwg. 8796522. The DOT designation shall be in accordance with U.S. Army ARDEC HCSD statement. Apply appropriate label.

5.3.3.1 Performance oriented packaging marking. Apply POP markings on side of box opposite nomenclature:

u 4C1/Y*/S/**
n USA/DOD/AYD

Note 1 - *1/2 the weight of pack in pounds, plus 1.
Note 2 - ** Date packaged.

5.3.3.2 Unitization. Unitization shall be in accordance with MIL-STD-1190.

5.4 Packaging, packing and marking for inert piston actuator parts. Packaging, packing and marking for inert piston actuator parts shall be in accordance with MIL-STD-1169 for all applicable levels of packaging.

5.5 Shipping. When piston actuators from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The Actuator, Piston, PA535 is intended for use in munition assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Distribution of ammunition data cards, inspection and test results, process control. Distribution of ammunition data cards, inspection and test process control shall be sent to:

Commander
AMCCOM
ATTN: AMSMC-QAR-R(D)
Picatinny Arsenal, New Jersey 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Subject term (key word) listing.
Actuator Piston
Shock G Test
Waterproofness

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A408)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70827 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:30Z
Check the source to verify that this is the current version before use.

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)